# Vamshi Bathini

Hanamkonda, Telangana, India | +91 9381618397 | vamshigoud7890@gmail.com
linkedin.com/in/bathini-vamshi-bb346a354 | github.com/Bathinivamshi | leetcode.com/u/vamshigoud7890

## Summary

Computer Science and Artificial Intelligence graduate with hands-on experience building full-stack and backend applications using Java, Spring Boot, React.js, and MySQL, plus machine learning services in Python. Skilled in REST APIs, Spring Security (JWT), JPA/Hibernate, SQL, Docker, and Git. Seeking a Software Engineer or AI/ML Engineer role.

## Technical Skills

**Languages:** Java, Python, C, C++, JavaScript, SQL
**Backend:** Spring Boot, Spring Security, REST APIs, JPA/Hibernate, JWT, Microservices, FastAPI
**Frontend:** React.js, HTML5, CSS3
**Databases:** MySQL
**AI/ML:** Machine Learning, Deep Learning, NLP, Scikit-learn, Random Forest, Logistic Regression, TF-IDF
**Tools & DevOps:** Git, GitHub, Docker, AWS, Maven, Gradle, Postman, Power BI
**CS Fundamentals:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## Projects

**AI-Powered Expense Tracker** | Java, Spring Boot, React.js, MySQL, Python, FastAPI

- Built a secure expense management REST API with Java, Spring Boot, Spring Security, JPA, and MySQL, including CRUD operations, validation, filtering, sorting, pagination, and JWT-based access control.
- Integrated an ML expense-categorization service (Python, Scikit-learn, TF-IDF, Logistic Regression) exposed through a FastAPI microservice with confidence-based category suggestions.
- Containerized the React frontend, Spring Boot backend, FastAPI ML service, and MySQL database with Docker for a multi-container build and deployment.

**AI-Powered Predictive Maintenance System** | Python, Scikit-learn, FastAPI, Streamlit

- Developed a Random Forest model on the UCI AI4I 2020 dataset to predict industrial machine failures, achieving 97.7% test accuracy and 95.9% ROC-AUC.
- Tuned the classification threshold to reach 61.8% failure recall, and built a FastAPI REST API and Streamlit dashboard for machine failure-risk prediction.

**Train Ticket Booking System** | Java, JDBC, MySQL, Gradle

- Built a Java booking application with user authentication, train search, seat booking, and cancellation, backed by persistent MySQL storage.
- Designed a layered architecture (Service, Repository, Entity) using SQL joins, foreign keys, prepared statements, and password hashing for secure data handling.

## Experience

**Machine Learning Intern (Virtual), AICTE Internship** | Jun 2024 - Jul 2024
*Google Developers Platform*

- Completed a virtual internship in Artificial Intelligence and Machine Learning through AICTE and Google Developers Platform.
- Practiced machine learning techniques, model development, and AI application building.

## Education

**SR University, Telangana, India** | Aug 2022 - Aug 2026
*Bachelor of Technology in Computer Science and Artificial Intelligence | CGPA: 8.1*

**Shivani Junior College, Hanamkonda** | Jul 2020 - Jul 2022
*Intermediate (MPC) | Percentage: 97.5%*

**Kakatiya Vidyalayam High School, Hanamkonda** | Jun 2010 - Jun 2020
*SSC | GPA: 10*

## Certifications

- Programming, Data Structures and Algorithms using Python - NPTEL
- Deep Learning for Developers - Infosys Springboard